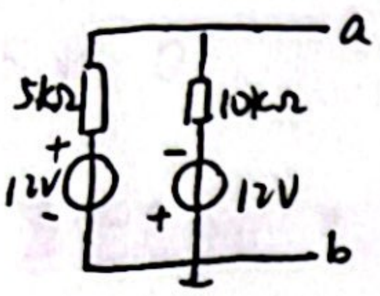
2019-2020

一. 填空题

(1) $\frac{d^2u_c}{dt^2}+10\frac{du_c}{dt}+5du_c=0$. 过阻尼.

(2) 导通 0.99mA

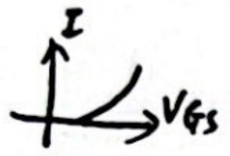
对左边戴维南等效：



$U_{ab}=12\times\frac{10}{10+5}-12\times\frac{5}{10+5}=4V$.

$R_{ab}=\frac{10}{3}k\Omega$.

$I_D=\frac{4-0.7}{10}mA=0.19mA$.

(3) b. 增强型 

(4) $(111100111)_2=(747)_8=(0100\,0111)_{8421BCD}$

(5) 6 001110

(6) $(110010111)_{补}=(10110100\,1)_{原}=(-105)_{10}$ 原码=补码自己补自己

(7) 0V

(8) 3.6 降低

(9) 是

(10) 8 2

二. 简答题

1. $Z=\bar{B}\bar{D}+A+C\bar{D}$ (写1)

$Z=\overline{\overline{B}\overline{D}+\overline{A}\overline{C}+\overline{A}\overline{D}}$ (写0取非)

2. $Z_1=\bar{A}\overline{AB}=\bar{A}$, $Z_2=\bar{B}$ (100Ω算小电阻自己话，我不太确定)

3.

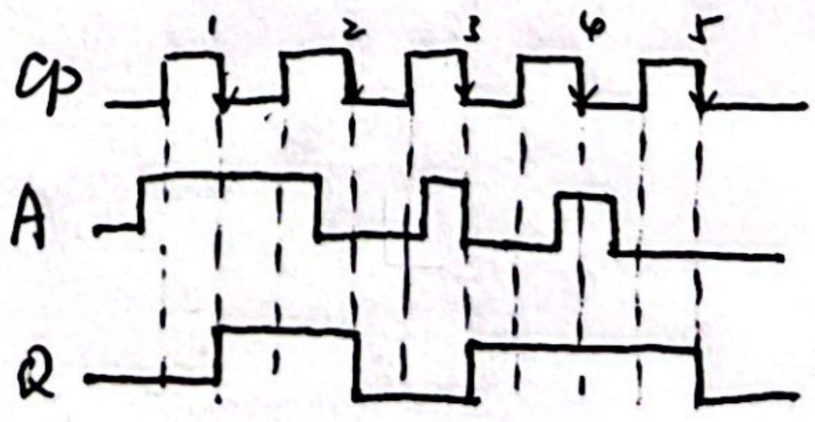
A	B	C	Z
0	0	×	0
0	1	0	0
0	1	1	1
1	0	×	0
1	1	×	1

卡诺图略

$Z=BC+AB$

4. $Q^{n+1}=J\bar{Q}^n+\bar{K}Q^n=\bar{Q}^n+A$

Cp	A	Q^n	Q^{n+1}
0		0	
1↓	1	×	1
2↓	0	1	0
3↓	×	0	1
4↓	1	1	1
5↓	0	1	0



5. 假设处于放大状态. $I_C=\beta I_B=50I_B$, $I_E=I_C+I_B=51I_B$.

$200I_B+0.7=15-3\times51I_B$.

$\Rightarrow I_B=\frac{15-0.7}{200+3\times51}=0.04mA$, $V_{CE}=-V_{CC}=-(15-2\times5)=-5V<-0.7V$

∴假设成立.

$I_C=50I_B=2mA$.

三. (1)

$A_3A_2A_1A_0$	Z
0 0 0 0	0
0 0 0 1	0
0 0 1 0	0
0 0 1 1	0
0 1 0 0	1
0 1 0 1	1
0 1 1 0	1
0 1 1 1	1
1 0 0 0	1
1 0 0 1	0

卡诺图略.

$Z=A_2+A_3\overline{A_0}$

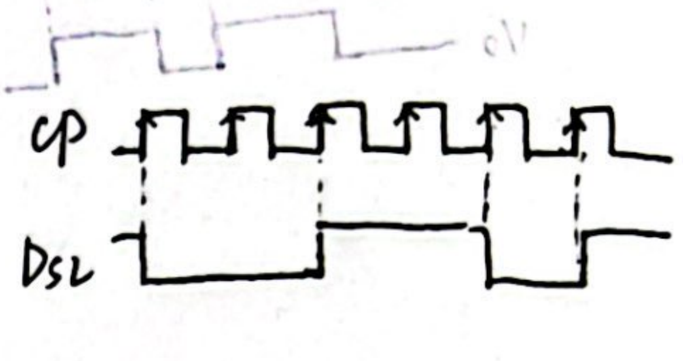
(2) $Z=\overline{\overline{A_2}\cdot\overline{A_3\overline{A_0}}}$

逻辑电路图略.

四.

CP	Q_3	$Q_2Q_1Q_0$	D_{SL}
0	0	1 1 0	1
1↑	1	1 0 1	0
2↑	1	0 1 0	0
3↑	0	1 0 0	1
4↑	1	0 0 1	1
5↑	0	0 1 1	0
6↑	0	1 0 0	1

$D_{SL}=\overline{\overline{Y_0}\,\overline{Y_4}\,\overline{Y_1}}=Y_0+Y_4+Y_1$



五.

$Q_1^nQ_0^n$	$Q_1^{n+1}Q_0^{n+1}$
0 0	1 0
1 0	0 1
0 1	0 0
1 1	0 0

$Q^{n+1}=J\overline{Q}^n+\overline{K}Q^n$ ↘

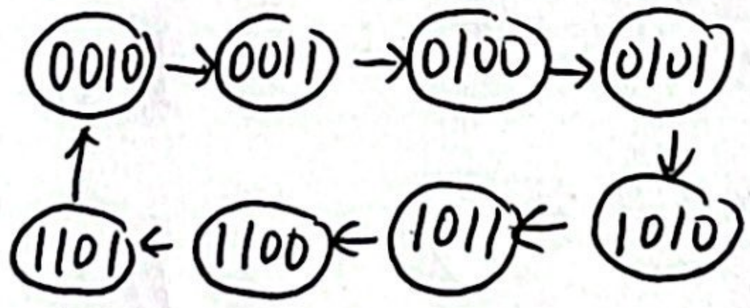
$Q_1^{n+1}=\overline{Q_1^n}\overline{Q_0^n}$, $J_1=\overline{Q_0^n}$, $K_1=1$

$Q_0^{n+1}=Q_1^nQ_0^n$, $J_0=Q_1^n$, $K_0=1$

连线画图略.

六. 1.

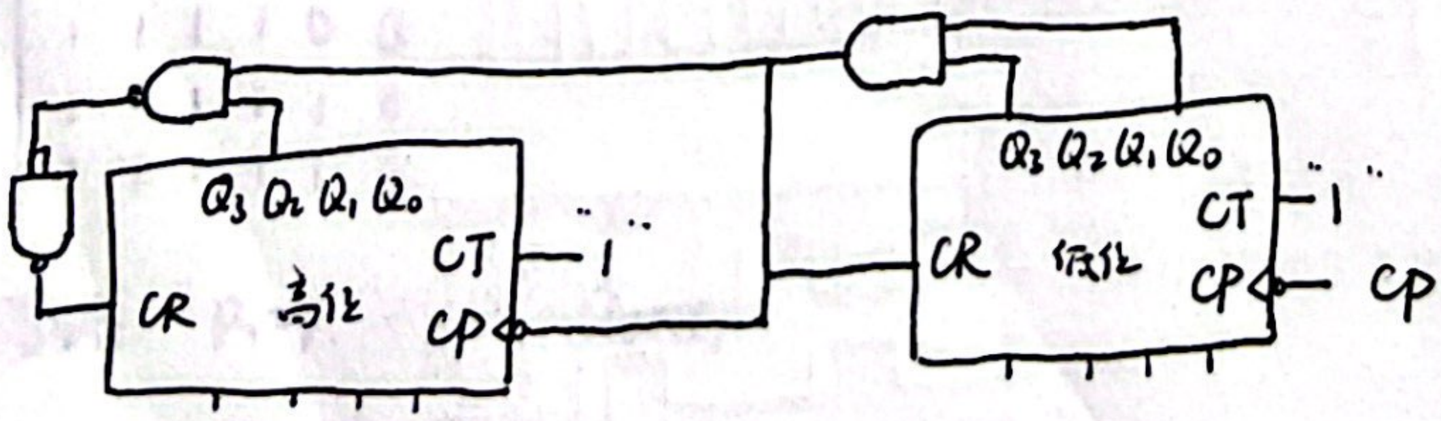
$Q_3^nQ_2^nQ_1^nQ_0^n$	$Q_3^{n+1}Q_2^{n+1}Q_1^{n+1}Q_0^{n+1}$
0 0 0 0	0 0 0 1
S	S
0 1 0 1	1 0 1 0
1 0 1 0	1 0 1 1
S	S
1 1 0 1	0 0 1 0
0 0 1 0	0 0 1 1
S	S
0 1 0 1	1 0 1 0
⋮	⋮



Q_2 00110011 $f_{cp_2}=200Hz$

Q_0 01010101 $f_{cp_0}=400Hz$.

2.

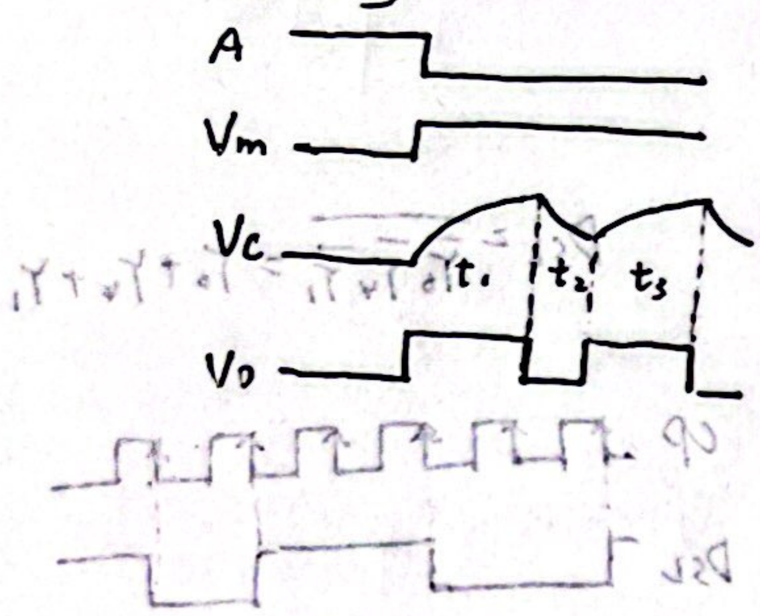


同为高电平清零.

七 (1) RS触发器

(2) 振荡电路.

(1) 如图



$t_1 = 20\times1\times\ln3 = 21.9ms$

$t_2 = 10\times1\times\ln2 = 6.9ms$

$t_3 = 20\times1\times\ln2 = 13.9ms$.

∴ $T_{WH} = 13.9ms$

$T_{WL} = 6.9ms$.